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COURSE NAME: DATA SCIENCE

BRIGHTLEARN RESEARCH ASSIGNMENT 2.

1. Business Analytics is when a company uses data to understand how it is doing and to make better decisions. It focuses on solving business problems using information from data. (Investopedia, 2024)

Example:

shoprite looks at sales data to decide which products to restock more often

2. Data Analytics is about looking at raw data to find patterns and trends. It focuses more on analysing the data itself, not always on making business decisions (Investopedia, 2024)

Example:

At Twinsover group, data from machines can be checked to see when breakdowns happen the most.

3. Business Intelligence (BI) is about using tools like dashboards and reports to show what is happening in a business. It mainly looks at historical and current data (Microsoft, 2023)

4. Data-Driven Decision Making means making decisions based on real data instead of ~~guess~~ assumptions or guessing. It helps companies to make better and more accurate choices (Harvard Business Review, 2012)

Example:

Woolworths may stop selling a product if the sales data shows it is not doing well.

5. Big Data is a very large amounts of data that come from different places or databases and grow very fast. It is usually too big to handle with normal tools (McKinsey, 2011)

Example:

checkers collect a lot of customer data from online and in-store shopping

6. Descriptive Analytics looks at historical data to show what has already occurred. It is usually shown in reports or charts (IBM, 2023)

Example:

SPAR creates monthly sales reports for each store

7. Diagnostic Analytics tries to explain why something happened. It looks deeper into the data to find causes (IBM, 2023)

Example!

Corruseal Group checks why production dropped by looking at machine issues.

8. Predictive Analytics uses past data to guess what might happen in the future. (SAS, 2023)

Example!

Shoprite predicts which products will sell more during holidays

9. Prescriptive analytics suggests what actions a company should take based on data (SAS, 2023)

Example!

Twinsave Group uses data to decide the best production schedule.

10. Data Literacy means being able to understand and use data properly. It helps people make better decisions (Qlik, 2020)

Example!

A supervisor at Corruseal Group reads reports to improve production

SECTION 2: DATA, MEASUREMENT & REPORTING

11. Structured data is data that is organised in tables with rows and columns. It is easy to read and analyse (IBM, 2023)

Example:

Production data in spreadsheets at TwinSaver Group.

12. Unstructured data is data that is not organised in a fixed format. It is harder to analyse (IBM, 2023)

Example:

Customer comments at Woolworths.

13. Data warehouse is a place where a company stores large amounts of data from different sources (Microsoft, 2023)

Example:

Shoprite store data from all stores in one system

14. Data Mining is when companies analyse large data sets to find patterns.

Example:

Checkers finds customer buying patterns

16. Dashboard shows important data in one place, often in real time (Microsoft, 2023)

Example:

SPAR uses dashboards to track store performance.

17. Metric is any number used to measure something, Not all metrics are important. (Investopedia, 2024)

Example:

Boxes produced at Corruseal Group.

18. KPI is an important metric that helps a business measure success (Hubspot, 2023)

Example:

Production efficiency at Twinsaver Group.

19. Benchmark is a standard used to compare performance against competitors (Investopedia, 2024)

Example:

Woolworths compares its performance or sales with Shoprite or Checkers.

20. Segmentation means dividing customers into groups (Hubspot, 2023)

Example:

Shoprite targets different customers into ~~groups~~ different promotions

SECTION 3: CUSTOMER CUSTOMER & SALES KPIs

21. Conversion rate is the percentage of people who take a desired action, like buying a product or signing up. It shows how successful a website or campaign is (Google Analytics, 2023)

Example:

If 200 people visit the Shoprite website and

20 place orders, the conversion rate is 10%.

22. Customer Acquisition Cost is the total cost a company spends to get one new customer. This includes marketing, advertising and sales costs. (Hubspot, 2023)

Example:

Abisa may spend money on online ads to attract new customers to open accounts. The total cost ~~divided~~ divided by number of new customers give CAC

23. Customer Lifetime Value (CLV/LTV) is the total amount of money a business expects to earn from a customer over the whole time they remain a customer. It shows how valuable a customer is in the long run. (Investopedia, 2024)

Example:

A loyal customer who shops regularly at Woolworths over several years has a high lifetime value.

CLV : CAC Ratio compares how much value a customer brings (CLV) to how much it costs to get them (CAC). A good business usually earns more from a customer than it spends to get them (Hubspot, 2023).

Example:

At checkers, if it costs R200 to get a customer

but they spend R800 over time, the ratio is 4:1, which is good.

24. Churn rate is the percentage of customers who stop buying or using a service over time. A high churn rate means the business is losing customers (Investopedia, 2024)

Example:

If customers stop shopping at SPAR, this increases the churn rate.

25. Customer retention rate is the percentage of customers who continue to buy from a business over time. It is the opposite of Churn (HubSpot, 2023)

Example:

Customers who keep returning to Woolworths increase the retention rate.

26. Net promoter score measures how likely customers are to recommend a business to others. It shows customer satisfaction and loyalty (HubSpot, 2024)

Example:

Checkers may ask customers if they would recommend the store to friends

27. Average order value (AOV) is the average amount of money a customer spends per ~~per~~ purchase (HubSpot, 2024)

Example:

Shoprite can calculate the average amount customers spend per shopping trip.

28. Sales Funnel shows the steps a customer goes through before making a purchase, from awareness to buying (Chaffey and Ellis-Chadwick, 2019).

Example:

Pick n Pay guides customers from ads to final purchase.

SECTION 4: MARKETING & CAMPAIGN ANALYTICS

29. Marketing Campaign is a planned set of activities used to promote a product or service over a certain period (Kotler and Keller, 2016).

Example:

MTN runs campaigns for data deals

30. Post-Campaign Analysis is when a company reviews a campaign after it ends to evaluate its success (McKinsey & Company, 2022)

Example:

Takealot checks if a campaign increased sales.

31. Incremental revenue is the extra income earned as a direct result of a campaign or action (McKinsey & Company, 2022)

Example:

shoprite earns additional revenue during promotions

32. Return on Investment (ROI) measures how much profit is made compared to the cost of an investment (Investopedia, 2024).

Example:

Standard Bank checks if marketing spending is profitable

33. Return on Ad Spend (ROAS) measures how much revenue is earned for every rand spent on advertising (Hubspot, 2024)

Example:

Checkers tracks revenue from online ads.

34. Click-Through rate (CTR) is the percentage of people who click on an ad after seeing it (Chaffey and Ellis-Chadwick, 2019).

Example:

fNB measures ad engagement online.

35. Bounce rate is the percentage of visitors who leave a website without interacting (Hubspot, 2024)

Example:

Takealot tracks users who leave without buying.

36. A/B Testing is when two versions of something are compared to see which performs better (Chaffey and Ellis-Chadwick, 2019).

Example:

Woolworths tests different adverts.

37. Lead Generation is the process of attracting and collecting potential customers (Kotler and Keller, 2016).

Example:

Capitel Bank collects details of interested clients.

38. Marketing Attribution identifies which channel led to a customer's purchase (McKinsey & Company, 2022)

Example:

Pick n Pay tracks whether ads or emails drove sales.

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